Size selection (Purification) (1X ratio)

Huang MEGA Lab (NCKU)
v1.3, 2026-10-21; pooled sample, 1X beads, 20 µL elution; hand-maintained since 2026-10-21

Source of record: Huang MEGA Lab (NCKU). Originally authored on protocols.io (dx.doi.org/10.17504/protocols.io.x54v9pez created 2026-09-13, version 1. This PDF is the lab's self-hosted copy and supersedes the protocols.io export.

Abstract

Size selection (Purification) with 1X ratio to keep more DNA than original version

Procedure

PROCEDURE

---

1. Note the volume of the **pooled sample** (the sum of the equal-molar volumes transferred into the pooling tube). Mix the bead bottle **VIGOROUSLY** so it is homogeneous, then pipette the **same volume of beads (1X)** into a new 1.5 mL tube (e.g. 180 µL pool → 180 µL beads).

Note Bead : sample = 1 : 1 (1X) keeps moderate to long DNA fragments. The clean-up is done once, on the pooled sample, not per tube.

Note **Magnetic bead**: BeaverBeads™ DNA Select Isolation

---

2. Add the beads to the pooled sample (**1 volume : 1 volume**).

Note **Pipetting thoroughly every time** before adding to make sure all magnetic beads mixed well.

---

3. Mix sample and beads by **gently flicking** the tube for **5 min**.

Note Be gentle, mix too aggressive will make DNA fragmented

---

4. Transfer the tube to the 8-strip magnetic rack.

After all the magnetic beads are arrested, remove the supernatant.

---

5. Add **200 µL** 80% ethanol, flip the magnetic rack around to clean the pellet.

Wait until all the magnetic beads are arrested, remove the supernatant.

Note **FRESHLY** make 80% ethanol

---

6. Repeat the **step 5**.

---

7. Flash spin the tube, and put on the magnetic rack.

Remove **ANY trace of liquid residuals** using 10 µl pipette.

Note Caution: DO NOT let the beads crack

---

8. Add **20 µL** elution buffer (17.5 µL of the eluate goes into the EvoPrep end-prep).

Mix gently by flicking the tube and make sure all magnetic beads are dissolved in buffer, then flash spin down

the sample.

9. Incubate the tube in PCR thermal cycler using "**37_incubation**" program for **10 min**.

10. Transfer the tube to the magnetic rack.
After all the magnetic beads are arrested, collect the **20 µL** supernatant into a 200 µL PCR tube.

11. The cleaned pool is ready for library construction (KAPA EvoPrep, 17.5 µL input).